

12312515 王路路 Assignment 3

Q1. $\underline{X} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \\ 1 & 2 \end{pmatrix}$, $\underline{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$, $\underline{\beta} = \begin{pmatrix} \theta \\ \varphi \end{pmatrix}$ Then we can get $\underline{\hat{\beta}} = (\underline{X}'\underline{X})^{-1}\underline{X}'\underline{y} = \begin{pmatrix} \frac{1}{6}(y_1 + 2y_2 + y_3) \\ \frac{1}{3}(-y_2 + 2y_3) \end{pmatrix}$

which means $\hat{\theta} = \frac{1}{6}(y_1 + 2y_2 + y_3)$, $\hat{\varphi} = \frac{1}{3}(-y_2 + 2y_3)$

Q2. Define the model: $\underline{y} = \underline{X}\underline{\beta} + \underline{\varepsilon}$, where $\underline{\beta} = (\theta, \varphi)'$

Then we can get: $\underline{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \\ y_{m+1} \\ \vdots \\ y_{m+n} \end{pmatrix}$, $\underline{X} = \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ \vdots & \vdots \\ 1 & -2 \\ \vdots & \vdots \end{pmatrix}$ Define $A = \frac{1}{m} \sum_{i=1}^m y_i$, $B = \frac{1}{m} \sum_{i=1}^m y_i$, $C = \frac{1}{n} \sum_{i=m+1}^{m+n} y_i$

and $\underline{X}'\underline{X} = \begin{pmatrix} 2m+n & m-2n \\ m-2n & m+4n \end{pmatrix}$, $(\underline{X}'\underline{X})^{-1} = \frac{1}{m^2+13mn} \begin{pmatrix} m+4n & -(m-2n) \\ -(m-2n) & 2m+n \end{pmatrix}$, $\underline{\hat{\beta}} = (\underline{X}'\underline{X})^{-1}\underline{X}'\underline{y} = \begin{pmatrix} \frac{Cm+An}{m+13n}A + \frac{Bn+3n}{m+13n}C \\ \frac{m(B-A)+n(2A+3B-5C)}{m+13n} \end{pmatrix}$

So $\hat{\theta} = \frac{Cm+An}{m+13n}$, $\hat{\varphi} = \frac{m(B-A)+n(2A+3B-5C)}{m+13n}$, $\text{Cov}(\hat{\theta}, \hat{\varphi}) = \frac{2n-m}{m^2+13mn} \sigma^2$ $\therefore \hat{\theta}$ and $\hat{\varphi}$ are uncorrelated if $m=2n$

Q3(a) $\tilde{\beta} - \beta = c\hat{\beta} - \beta = c(\hat{\beta} - \beta) + (c-1)\beta$ $\therefore \text{MSE}(\tilde{\beta}) = E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)'] = E[c(\hat{\beta} - \beta) + (c-1)\beta][c(\hat{\beta} - \beta) + (c-1)\beta]'$

Since $E(\hat{\beta}) = \beta$, $\text{MSE}(\tilde{\beta}) = c^2 E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] + (c-1)^2 \beta\beta'$ where $E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] = \text{tr}(\text{Cov}(\hat{\beta})) = \sigma^2(\underline{X}'\underline{X})^{-1}$

Then $\text{MSE}(\tilde{\beta}) = c^2 \sigma^2 \text{tr}(\underline{X}'\underline{X})^{-1} + (c-1)^2 \beta\beta'$

(b) $\frac{d}{dc} \text{MSE}(\tilde{\beta}) = 2c \sigma^2 \text{tr}(\underline{X}'\underline{X})^{-1} + 2(c-1)\beta\beta' = 0 \Rightarrow c^* = \frac{\beta\beta'}{\beta\beta' + \sigma^2 \text{tr}(\underline{X}'\underline{X})^{-1}} \leq 1$

(c) $\beta\beta' = 55$, $\text{tr}(\underline{X}'\underline{X})^{-1} = \frac{137}{60}$ $\therefore c^* = \frac{55}{55 + \frac{137}{60}} = \frac{3300}{3637}$

Q4 (a) $\frac{\partial}{\partial \underline{\beta}} (\underline{Y} - \underline{X}\underline{\hat{\beta}})'(\underline{Y} - \underline{X}\underline{\hat{\beta}}) = 0 \Rightarrow \underline{X}'\underline{X}\underline{\hat{\beta}} = \underline{X}'\underline{Y} \Rightarrow (\underline{M}_1 + \underline{X}_2'\underline{X}_2)\underline{\hat{\beta}} = \underline{X}_1'\underline{Y}_1 + \underline{X}_2'\underline{Y}_2$ Similarly: $\underline{X}_1'\underline{X}_1\underline{\hat{\beta}}^* = \underline{X}_1'\underline{Y}_1$

Then we can get: $\underline{M}_1(\underline{\hat{\beta}} - \underline{\hat{\beta}}^*) = \underline{X}_2'(\underline{Y}_2 - \underline{X}_2\underline{\hat{\beta}}) = \underline{X}_2'\underline{e}_2 \therefore \underline{\hat{\beta}} - \underline{\hat{\beta}}^* = \underline{M}_1^{-1}\underline{X}_2'\underline{e}_2$

Since $\underline{M}_1$ is full rank.

(b) $\underline{e}_2 = \underline{Y}_2 - \underline{X}_2\underline{\hat{\beta}} = \underline{Y}_2 - \underline{X}_2\underline{\hat{\beta}} + \underline{X}_2\underline{\hat{\beta}}^* - \underline{X}_2\underline{\hat{\beta}}^* = \underline{Y}_2 - \underline{X}_2\underline{\hat{\beta}}^* - \underline{X}_2(\underline{\hat{\beta}} - \underline{\hat{\beta}}^*) = \underline{e}_2^* - \underline{X}_2(\underline{\hat{\beta}} - \underline{\hat{\beta}}^*)$

$\Rightarrow \underline{\hat{\beta}} - \underline{\hat{\beta}}^* = \underline{M}_1^{-1}\underline{X}_2'(\underline{e}_2^* - \underline{X}_2(\underline{\hat{\beta}} - \underline{\hat{\beta}}^*)) \Rightarrow \underline{\hat{\beta}} - \underline{\hat{\beta}}^* = (\underline{M}_1 + \underline{X}_2'\underline{X}_2)^{-1}\underline{X}_2'\underline{e}_2^*$

(c) $\underline{M}_1 = \underline{X}_1'\underline{X}_1 = \begin{pmatrix} 7 & 0 \\ 0 & 28 \end{pmatrix}$, $\underline{\hat{\beta}}^* = (6, -2)'$, $\underline{X}_2 = (1 \ 4)$, $\underline{Y}_2 = 4$, $\underline{y}^* = 6 + (-2) \times 4 = -2$, $\underline{e}_2^* = 6$

$\therefore \underline{M}_1 + \underline{X}_2'\underline{X}_2 = \begin{pmatrix} 8 & 4 \\ 4 & 44 \end{pmatrix}$, $(\underline{M}_1 + \underline{X}_2'\underline{X}_2)^{-1} = \frac{1}{336} \begin{pmatrix} 44 & -4 \\ -4 & 8 \end{pmatrix}$

$\therefore \underline{\hat{\beta}} - \underline{\hat{\beta}}^* = (\underline{M}_1 + \underline{X}_2'\underline{X}_2)^{-1}\underline{X}_2'\underline{e}_2^* = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$, $\underline{\hat{\beta}} = \begin{pmatrix} 6.5 \\ -1.5 \end{pmatrix}$